



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination Dec.2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science) / B.Tech. CSE (Hons) in Artificial Intelligence and Machine Learning / B.Tech. CSE (Hons) in Cyber Security
Semester/Year:	Semester- 3
Paper Code:	BTECSDS301T22
Title of Paper:	Discrete Mathematical Structure
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section A and Section B. Section-A consists of seven questions and Section-B consists of six question. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question

Number

Question

Marks

Allotted

Section-A

Attempt any five questions.

- Q.1.** If $A = \{3, 6, 9, 12, 15, 18, 21\}$, $B = \{4, 8, 12, 16, 20\}$, $C = \{2, 4, 6, 8, 10, 12, 14, 16\}$, $D = \{5, 10, 15, 20\}$; find (4)
- (i) $A - B$ (ii) $A - C$
(iii) $A - D$ (iv) $B - A$

- Q.2.** Answer the following questions concerning the poset $(X, /(\text{divides}))$, where $X = \{2, 4, 6, 9, 12, 18, 27, 36, 48, 60, 72\}$, (4)
- (i) Find the maximal and minimal element.
(ii) Find the greatest and least element, if it exists.
(iii) Find lub of $\{2, 9\}$, if it exist.
(iv) Find glb of $\{60, 72\}$, if it exist.

- Q.3. Show that the set of even integers is a sub semi group but set of odd integers is not a sub semi group of the set of Integers for addition. (4)
- Q.4. Show that the set $\langle S = \{1, 2, 3, 6, 12\}, * \rangle$ is a monoid, where $a * b = \text{greatest common divisor of } a \text{ and } b$. Also find the identity element. (4)
- Q.5. Show that the implication $p \rightarrow (q \vee \neg r)$ for propositions p, q, r is truth functionally contingent. (4)
- Q.6. Explain the Bipartite and complete bipartite graph with example. (4)
- Q.7. Find r if: ${}^5P_r = 2^6 P_{r-1}$. (4)

Section-B

Attempt any four questions.

- Q.1. Show that in the power-set $P(A)$ of all subsets of a set A , set inclusion, $\subseteq$, is a partial order relation. (10)
- Q.2. Prove that the relation of 'Congruency modulo n ' defined as aRb if $a \cong b \pmod{n}$ i.e. when $a - b = \text{a multiple of } n$, is an equivalence relation in the set of integers I (10)
- Q.3. Show that the set G of four mappings $G = \{f_1, f_2, f_3, f_4\}$ defined by $f_1(x) = x$, $f_2(x) = -x$, $f_3(x) = \frac{1}{x}$, $f_4(x) = \frac{-1}{x}$, for $x \neq 0$, is a group under the 'operation composition of mappings'. (10)
- Q.4. From the truth tables of the statement solve the following: (10)
- | | |
|---|--|
| (i) $(p \rightarrow q) \vee (\neg p \rightarrow q)$ | (ii) $(p \rightarrow q) \wedge (\neg p \rightarrow q)$ |
| (iii) $(p \leftrightarrow q) \vee (\neg p \rightarrow r)$ | (iv) $(p \leftrightarrow q) \wedge (\neg p \rightarrow r)$ |
- Q.5. (a) Determine which of the following can be a degree sequence of a possible graph. (10)
- | | |
|-------------------|-----------------------|
| (i) 0, 2, 2, 3, 4 | (ii) 1, 1, 2, 3 |
| (iii) 2, 2, 4, 6 | (iv) 2, 2, 3, 4, 5, 5 |
- (b) Find the size of a r -regular (p, q) graph.
- (c) Define binary search tree.
- (d) Construct a 3-regular graph of 12 vertices.
- (e) Draw the complete bipartite graphs $K_{3,3}$ and $K_{4,5}$.
- Q.6. Solve the recurrence relation $a_r - 4a_{r-1} + 4a_{r-2} = 2^r$. (10)